

Senior Project Quarter Plan

Project: AI Powered Nutrition & Ayurveda App

Project Goal

Develop a web-based application that helps users make everyday food decisions using a combination of Ayurvedic principles, modern nutrition context, and AI-generated guidance. The app prioritizes non-judgmental, balanced recommendations rather than strict tracking or dieting.

Timeline & Milestones

Weeks 1–2: Product Definition & UI Foundation

- Finalize Product Requirements Document (PRD)
- Design core user flows (Home, Pantry Helper, Profile)
- Implement initial frontend using React + TypeScript + Tailwind
- Build static UI for all screens

Outcome: Functional UI prototype with navigation and core structure

Weeks 3–4: Core Logic (Rule-Based System)

- Implement rule-based logic for:
 - Time-of-day dosha cycles
 - Basic digestion/stress context handling
 - Pantry input → structured interpretation
- Create a “Pantry Helper” system that generates suggestions using deterministic logic
- Refactor code into modular structure (`/lib` for logic)

Outcome: App produces dynamic (non-static) suggestions without AI

Weeks 5–6: AI Integration

- Add backend endpoint for AI-generated responses
- Integrate LLM (OpenAI API) for:

- Natural language input interpretation
 - Generating meal suggestions and explanations
- Ensure structured JSON output from AI for frontend compatibility

Outcome: Pantry Helper transitions from rule-based to AI-assisted recommendations

Weeks 7–8: Refinement & Evaluation

- Improve UX (loading states, clarity, readability)
- Add disclaimers and ensure responsible guidance
- Conduct informal user testing (2–5 users)
- Evaluate:
 - usability
 - clarity of recommendations
 - perceived usefulness

Outcome: Polished prototype + qualitative feedback

Week 9–10: Finalization

- Final UI polish and bug fixes
- Prepare final demo
- Write final report:
 - product decisions
 - technical architecture
 - tradeoffs (rules vs AI)
 - future improvements

Outcome: Completed project + presentation-ready system

Technical Approach

- **Frontend:** React, TypeScript, Tailwind CSS
- **Logic Layer:** Rule-based Ayurvedic heuristics (time, digestion, stress)
- **AI Layer:** LLM for natural language understanding and response generation
- **Architecture:** Hybrid system (rules + AI, not fully AI-dependent)